

Celebal Excellence Summer Internship Program 2026

Week 4 Assignment Report: Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Submitted By: Manjit Bajaj

Domain: Data Engineering

StudentId: CT_CSI_DE_1160

Objective

The objective of this assignment was to understand the basic concepts of Microsoft Azure Cloud and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The assignment included creating cloud resources, storing data in Azure Blob Storage, configuring Azure Data Factory, creating a data pipeline using Get Metadata and Copy Data activities, executing the pipeline, monitoring the execution, and assigning IAM roles to securely access Azure Storage.

Tools and Services Used

- Microsoft Azure Portal
- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory (ADF)
- Azure IAM (Identity and Access Management)
- Superstore CSV Dataset

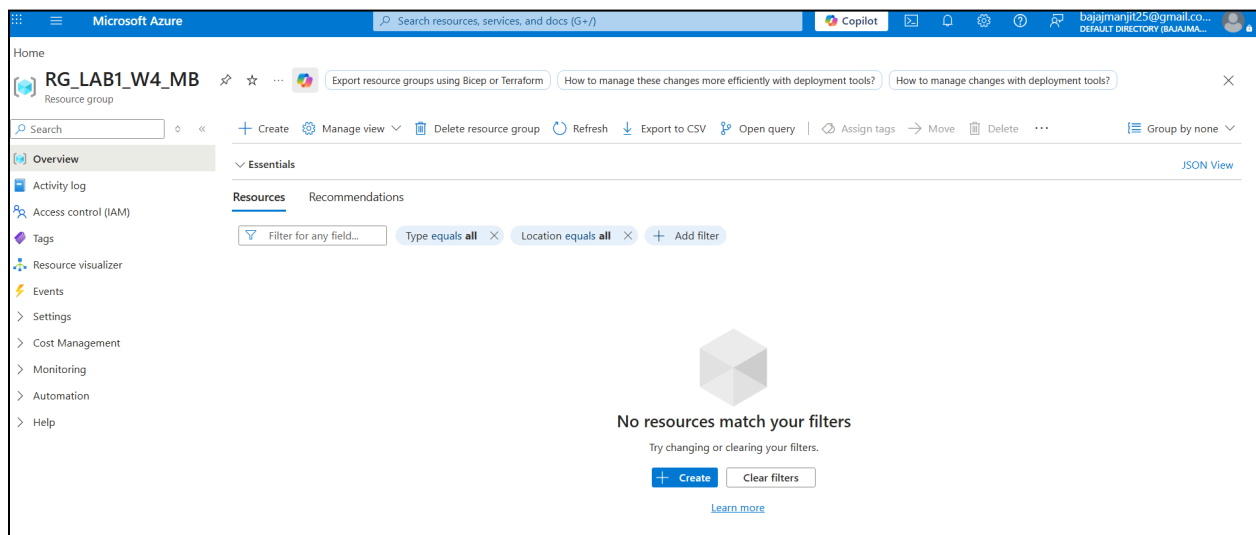
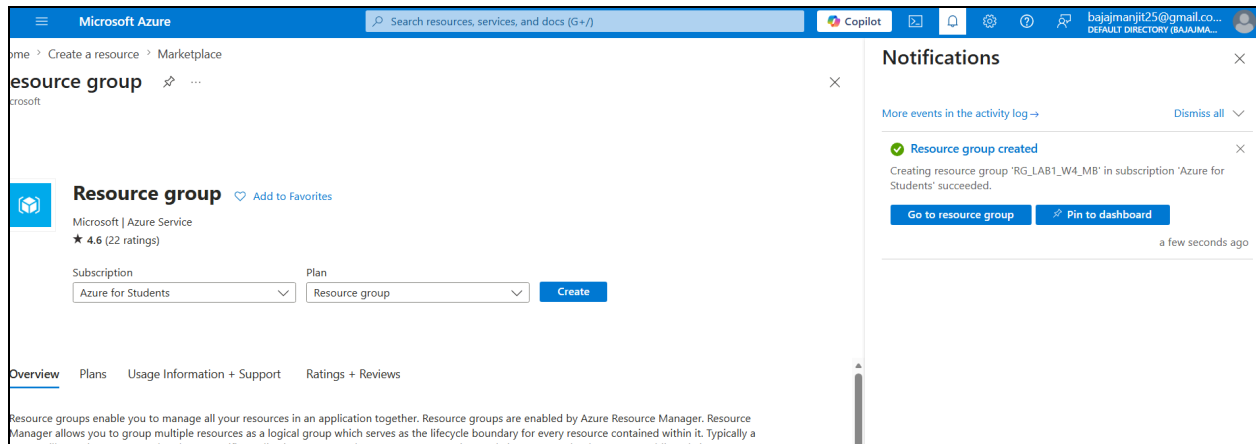
Task 1: Azure Resource Group Creation

Description

A Resource Group was created in Azure Portal to organize and manage the resources used for this project.

Steps Performed

1. Logged into the **Azure Portal**.
2. Opened **Resource Groups** from the Azure search bar.
3. Clicked on **Create**.
4. Selected the required subscription.
5. Entered the Resource Group name: **RG_LAB1_W4_MB**.
6. Selected the region: **Central India**.
7. Clicked **Review + Create**.
8. Successfully created the Resource Group in Azure Portal.



Task 2: Azure Storage Setup

Description

A Storage Account was created in Azure Portal to store and manage project data. A Blob Container was created inside the Storage Account, and a CSV file was uploaded to the container.

Steps Performed

Step 1: Create a Storage Account

1. Logged into the **Azure Portal**.
2. Searched for **Storage Accounts**.
3. Clicked **Create** to create a new Storage Account.
4. Selected the subscription:

- **Azure for Students**
- 5. Selected the existing Resource Group:
 - **RG_LAB1_W4_MB**
- 6. Entered the Storage Account name:
 - **mbstorage2026** (or your created storage account name)
- 7. Selected the region:
 - **Central India**
- 8. Configured the storage settings:
 - **Performance:** Standard
 - **Replication:** Locally Redundant Storage (LRS)
 - **Access Tier:** Hot
- 9. Clicked **Review + Create**.
- 10. Successfully created the Storage Account.

Microsoft Azure

Home > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. [Learn more about Azure storage accounts](#)

Project details

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * Azure for Students

Resource group * RG_LAB1_W4_MB [Create new](#)

Instance details

Storage account name * mbstorage2026

Region * (Asia Pacific) Central India [Deploy to an Azure Extended Zone](#)

Primary service * Azure Blob Storage or Azure Data Lake Storage

Performance * Standard: Recommended for most scenarios (general-purpose v2 account)

Microsoft Azure

Home > Storage center | Blob Storage

Create a storage account

Basics Advanced Networking Data protection Security Encryption Tags Review + create

[View automation template](#)

Basics

Subscription	Azure for Students
Resource group	RG_LAB1_W4_MB
Location	East Asia
Storage account name	mbstorage2026
Primary service	Azure Blob Storage or Azure Data Lake Storage
Performance	Standard
Replication	Locally redundant storage (LRS)

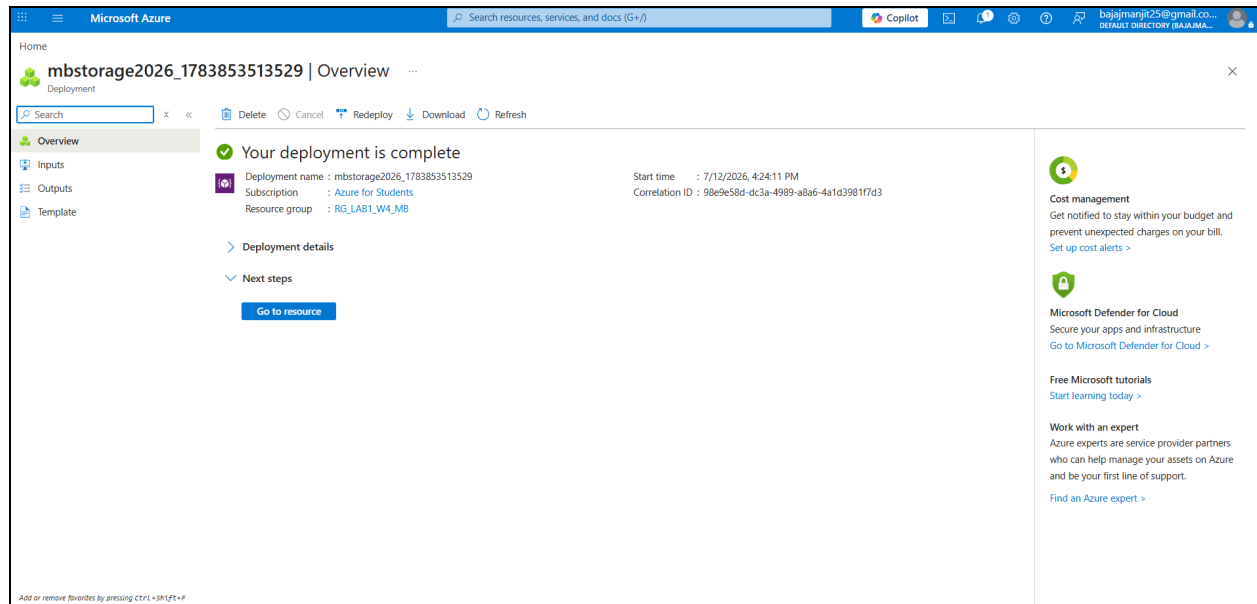
Advanced

Enable hierarchical namespace	Disabled
Enable SFTP	Disabled
Enable network file system v3	Disabled
Allow cross-tenant replication	Disabled
Access tier	Hot
Managed Identity for SMB	Disabled
Require Encryption in Transit for SMB	Enabled

Security

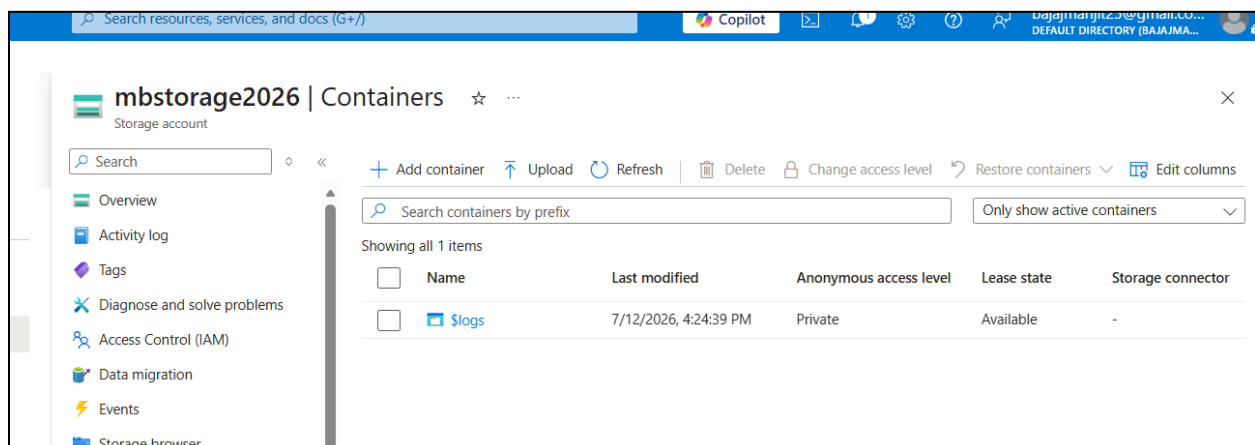
Previous Next **Create**

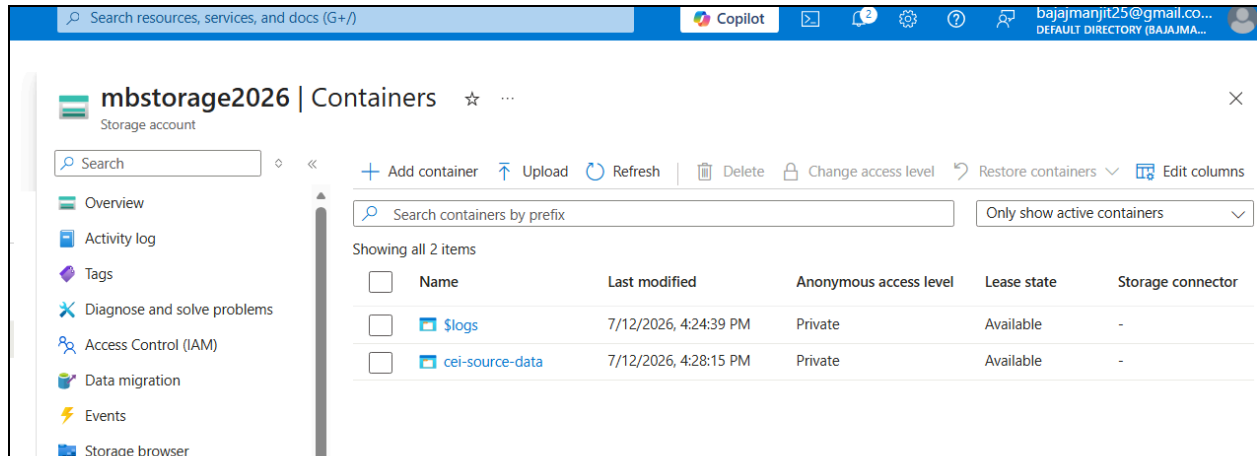
[Give feedback](#)



Step 2: Create a Blob Container

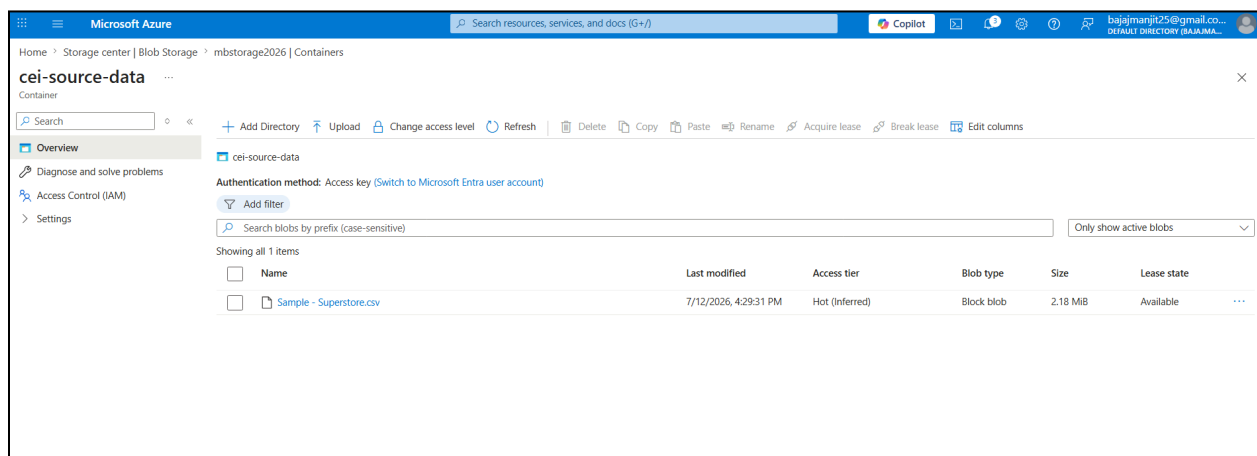
1. Opened the created **Storage Account**.
2. Navigated to:
 - **Data Storage → Containers**
3. Clicked + **Container**.
4. Entered the container name:
 - **cei-source-data**
5. Selected the access level:
 - **Private (No anonymous access)**
6. Clicked **Create**.
7. Successfully created the Blob Container.





Step 3: Upload a CSV File

1. Opened the created Blob Container.
2. Clicked the **Upload** button.
3. Selected the **Superstore dataset CSV file** from the local system.
4. Uploaded the file successfully.
5. Verified that the CSV file was visible inside the Blob Container.



Task 3: Azure Data Factory (ADF) Basics

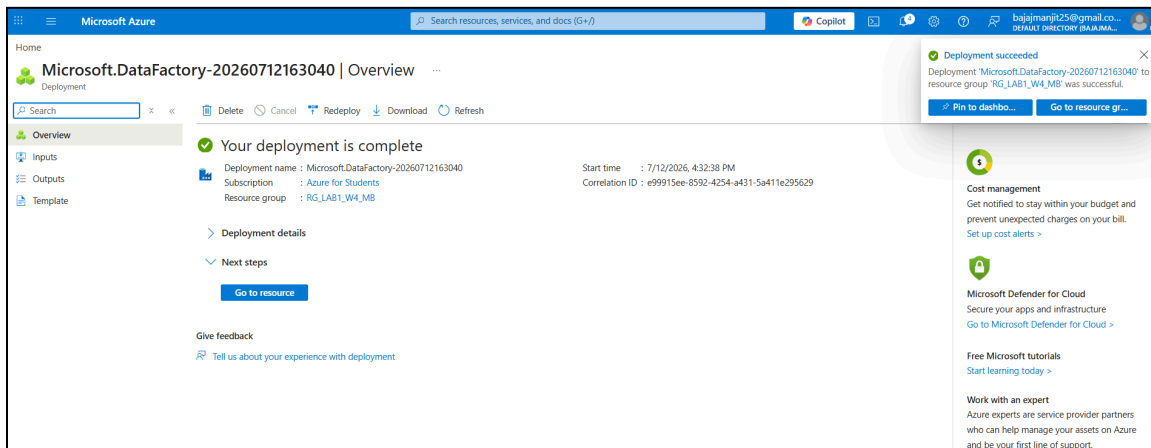
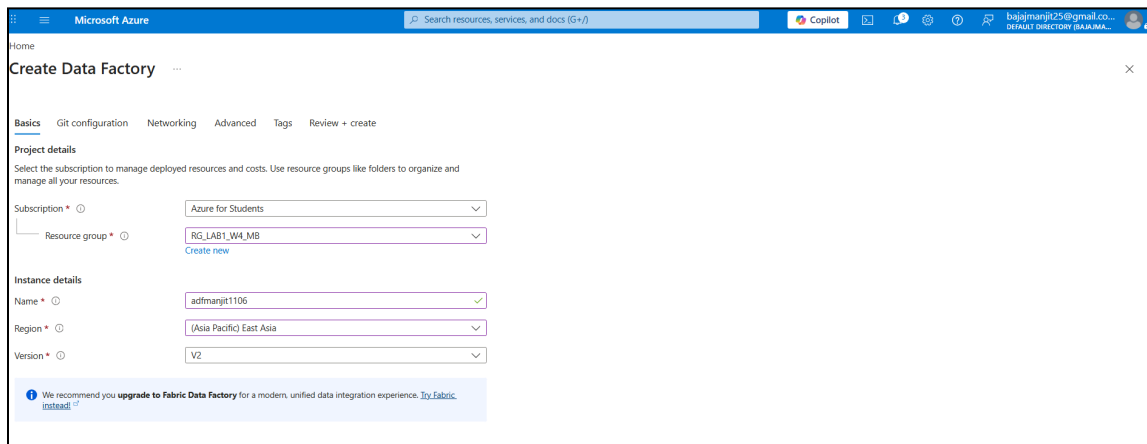
Description

Azure Data Factory (ADF) was created to build and manage data integration workflows. A Linked Service was created to connect Azure Data Factory with Azure Blob Storage. Datasets were created for source and destination data, and the Get Metadata activity was used to retrieve information about the data files.

Steps Performed

Step 1: Create Azure Data Factory

1. Logged into the **Azure Portal**.
2. Searched for **Data Factories**.
3. Clicked **Create** to create a new Azure Data Factory.
4. Selected the subscription:
 - **Azure for Students**
5. Selected the Resource Group:
 - **RG_LAB1_W4_MB**
6. Entered the Data Factory name:
 - **adfmanjit1106**
7. Selected the region:
 - **Central India**
8. Clicked **Review + Create**.
9. Successfully created the Azure Data Factory.



Step 2: Explore ADF User Interface

After creating the Data Factory, the ADF Studio interface was opened and the following sections were explored:

Author

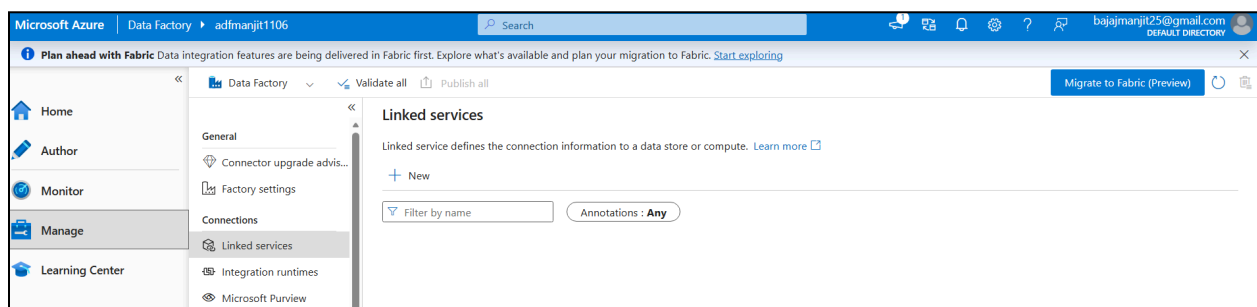
- Used to create and design data pipelines.
- Allows creation of activities, datasets, and linked services.

Monitor

- Used to monitor pipeline execution.
- Displays pipeline runs, activity status, and execution details.

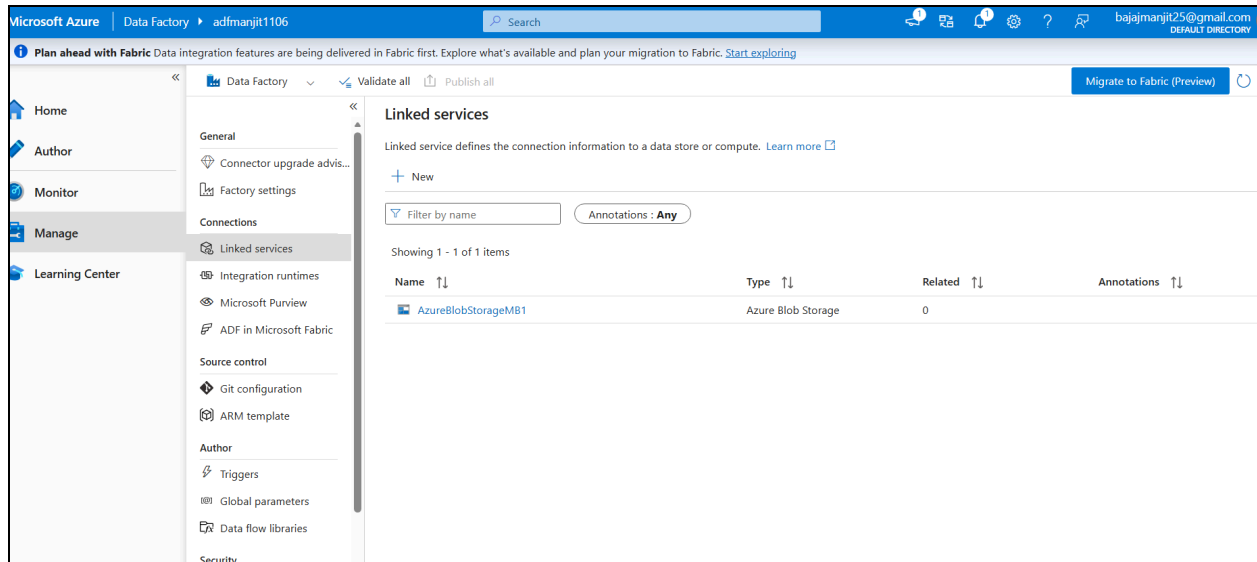
Manage

- Used to manage connections, triggers, linked services, and integration runtimes.



Step 3: Create Linked Service (Blob Storage)

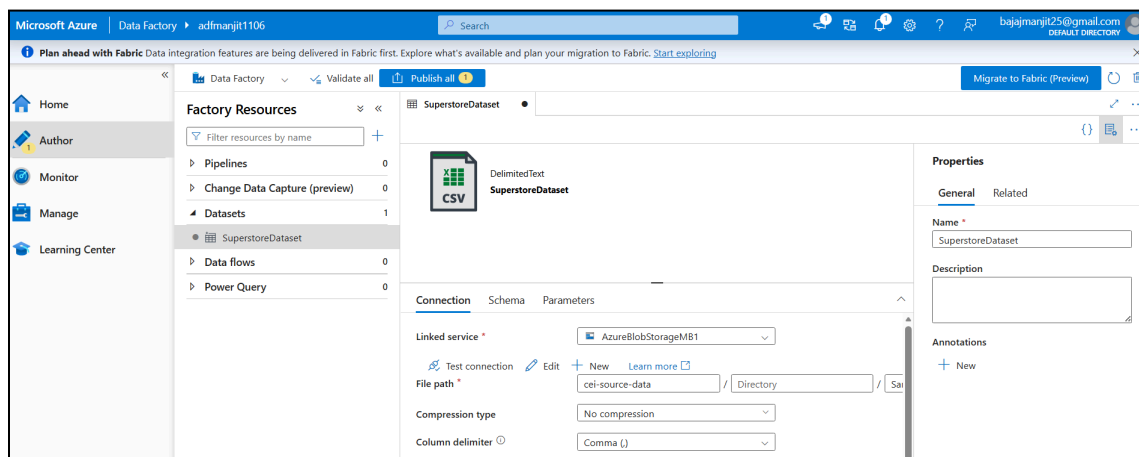
1. Opened **Azure Data Factory Studio**.
2. Navigated to:
 - **Manage → Linked Services**
3. Clicked **New Linked Service**.
4. Selected:
 - **Azure Blob Storage**
5. Entered the connection details:
 - Storage Account: **mbstorage2026**
6. Tested the connection.
7. Clicked **Create**.
8. Successfully created the Blob Storage Linked Service.



Step 4: Create Datasets (Source and Destination)

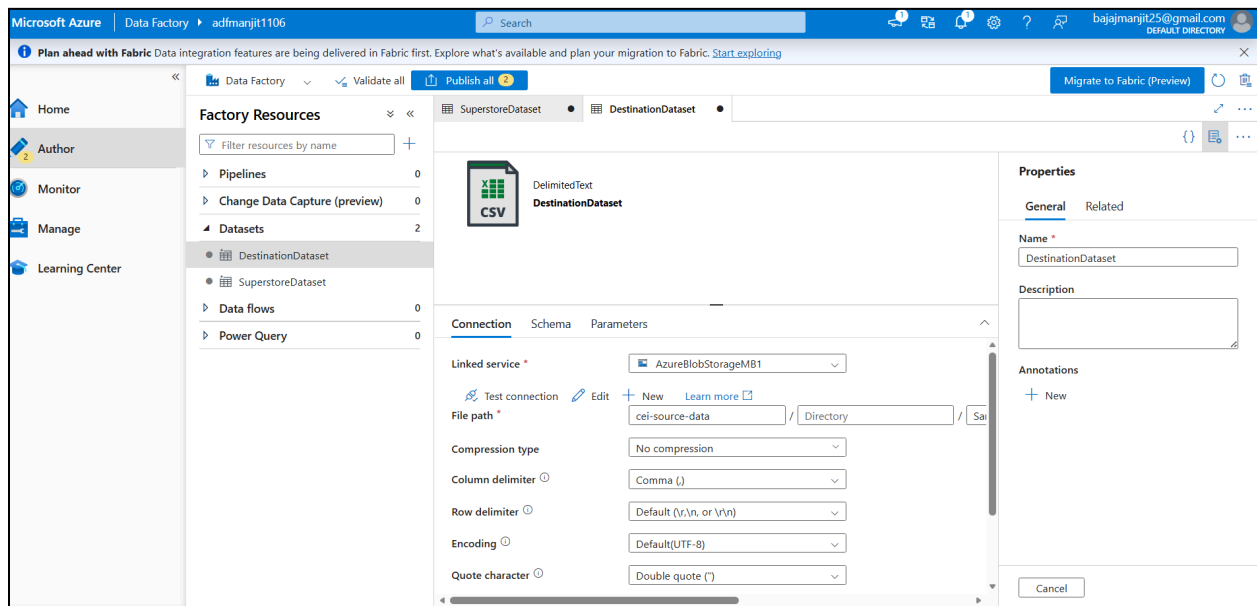
Source Dataset

1. Opened the **Author** section.
2. Selected **Datasets** → **New Dataset**.
3. Selected:
 - **Azure Blob Storage**
4. Chose the file format:
 - **Delimited Text (CSV)**
5. Connected it with the Blob Storage Linked Service.
6. Selected the uploaded CSV file as the source dataset.
7. Created the source dataset.



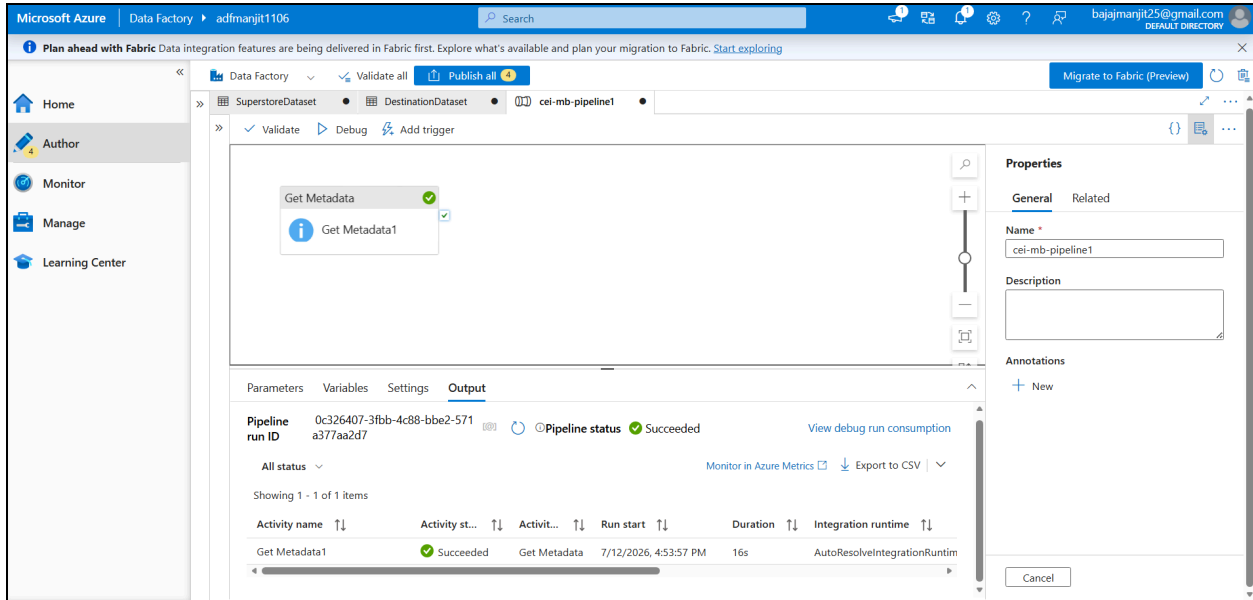
Destination Dataset

1. Created another dataset.
2. Selected:
 - **Azure Blob Storage**
3. Selected:
 - **Delimited Text (CSV)**
4. Configured the destination location.
5. Created the destination dataset.



Step 5: Use Get Metadata Activity

1. Opened the **Author** section.
2. Created a new pipeline.
3. Added the **Get Metadata** activity from the Activities panel.
4. Connected the source dataset with the Get Metadata activity.
5. Configured metadata fields such as:
 - File name
 - File size
 - Last modified date
6. Published the pipeline.



Task 4: Pipeline Development

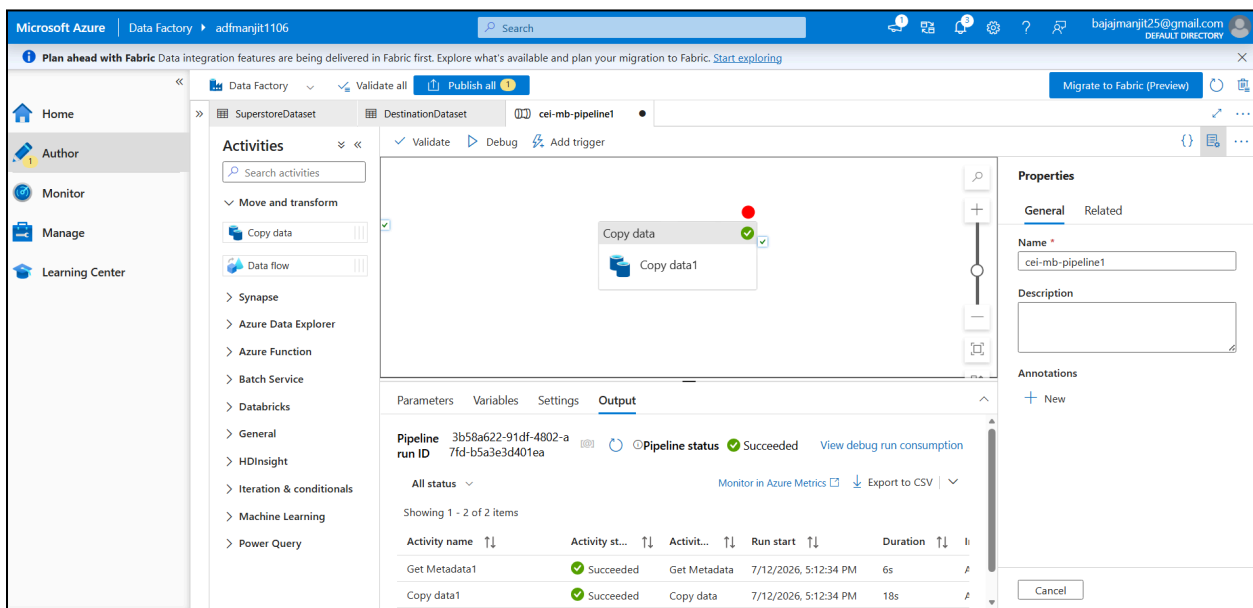
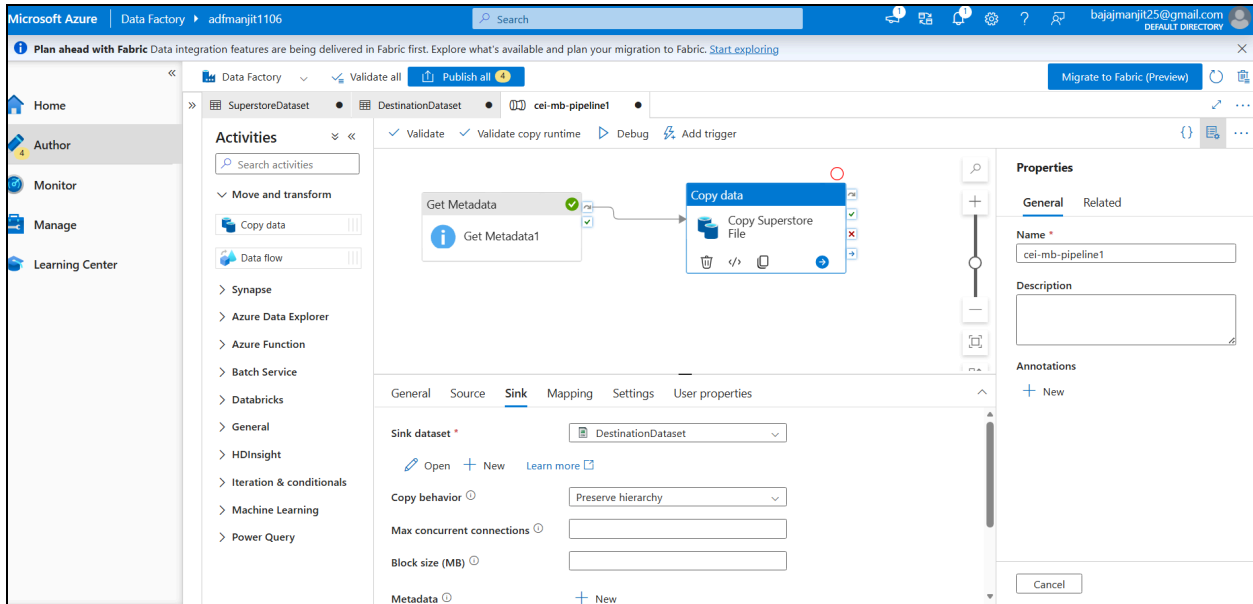
Description

An Azure Data Factory pipeline was created to copy data from the source location to the destination location using the **Copy Data activity**. The pipeline was configured with source and destination datasets to perform data movement.

Steps Performed

Step 1: Create Pipeline Using Copy Data Activity

1. Opened **Azure Data Factory Studio**.
2. Navigated to the **Author** section.
3. Clicked **New Pipeline**.
4. Added the **Copy Data activity** from the Activities panel.
5. Configured the Copy Data activity:
 - Selected the **Source dataset** (CSV file from Blob Storage).
 - Selected the **Destination dataset** (target Blob Storage location).
6. Configured source and destination connections using the Blob Storage Linked Service.
7. Validated the pipeline configuration.
8. Published the pipeline successfully.



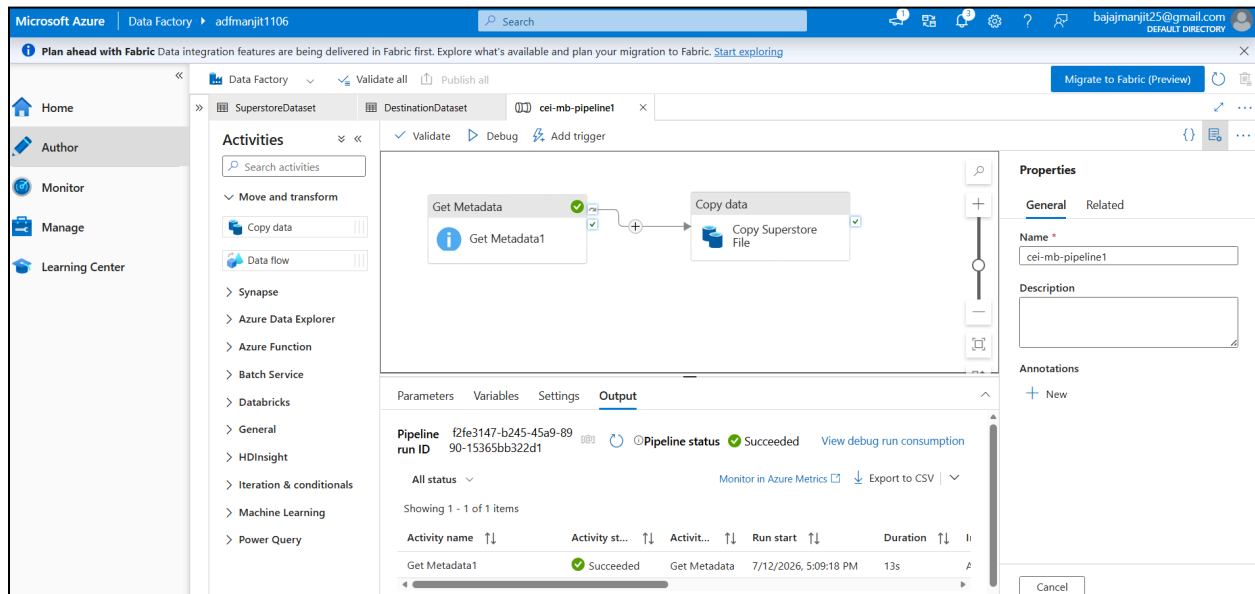
Task 5: Pipeline Execution

Description

The Azure Data Factory pipeline was executed to verify that the data copy operation was working successfully.

Steps Performed

1. Opened the created pipeline in **Azure Data Factory Studio**.
2. Clicked **Debug** to test the pipeline execution.
3. Monitored the pipeline run status.
4. Checked the execution details in the **Monitor** section.
5. Verified that the pipeline execution status changed to **Succeeded**.



Task 6: IAM Roles

Description

Azure Identity and Access Management (IAM) roles were assigned to control permissions for Azure resources. Reader and Contributor roles were configured, and Azure Data Factory was provided access to the Storage Account.

Steps Performed

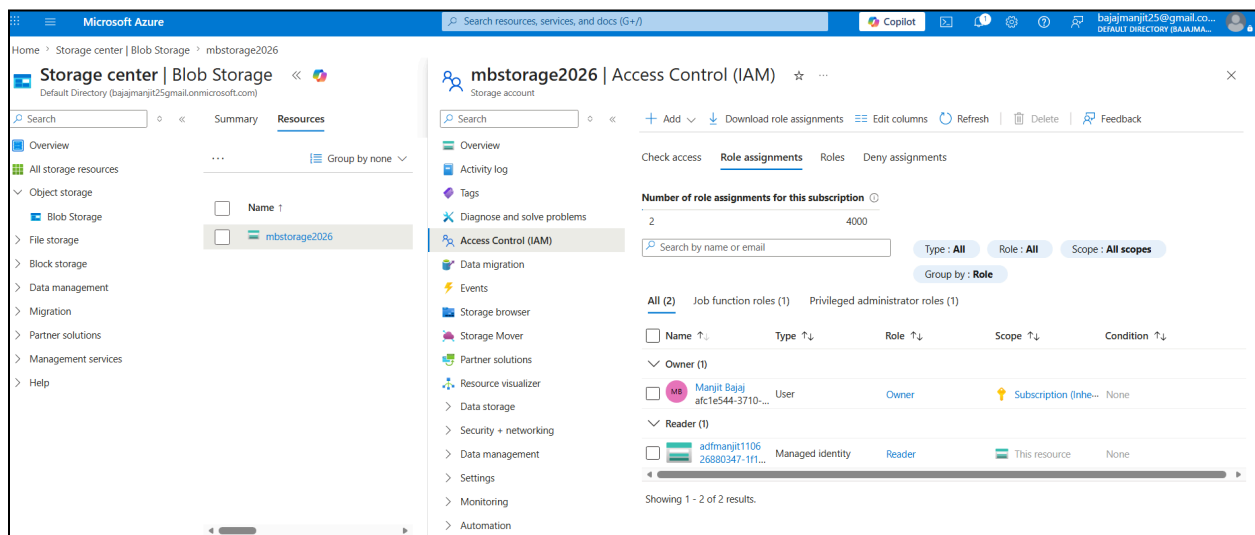
Step 1: Assign Reader Role

1. Opened the Azure Storage Account.
2. Navigated to:
 - **Access Control (IAM)**
3. Clicked:
 - **Add → Add Role Assignment**
4. Selected role:
 - **Reader**

5. Selected the required user/service.
6. Clicked **Review + Assign**.

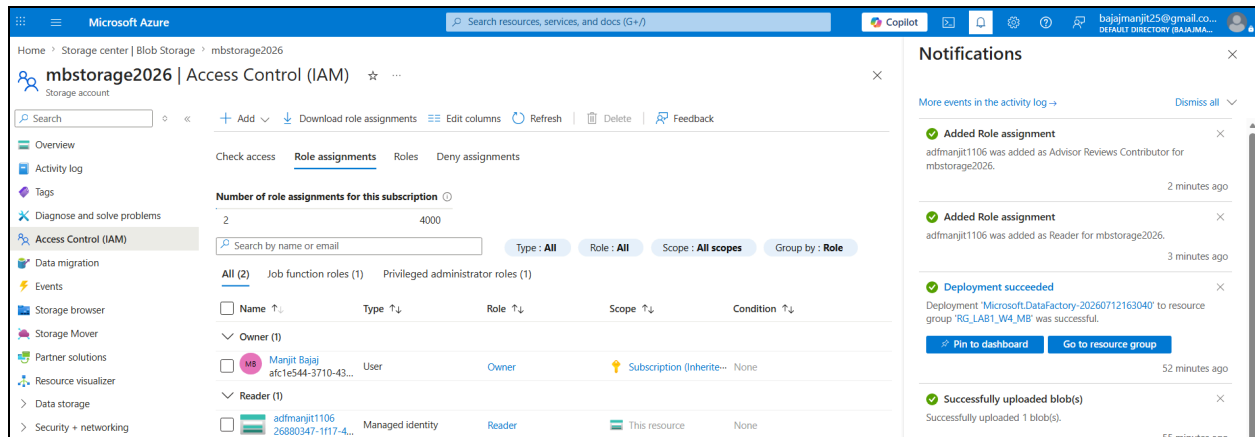
Step 2: Assign Contributor Role

1. Opened **Access Control (IAM)** again.
2. Clicked:
 - **Add Role Assignment**
3. Selected role:
 - **Contributor**
4. Assigned the role to the required account/service.
5. Clicked **Review + Assign**.



Step 3: Provide ADF Access to Storage

1. Opened the Storage Account.
2. Went to:
 - **Access Control (IAM)**
3. Added the Azure Data Factory identity.
4. Assigned the required storage access role:
 - **Storage Blob Data Contributor**
5. Saved the role assignment.



Final Outcome

The Azure Data Factory solution was successfully configured with:

- Storage Account integration
- Linked Services and datasets
- Copy Data pipeline
- Pipeline execution monitoring
- IAM-based access control for secure resource management.

Conclusion

This assignment successfully demonstrated the implementation of an end-to-end data pipeline using Microsoft Azure services. Azure Blob Storage was used to store the dataset, while Azure Data Factory was used to retrieve metadata and copy the file from the source container to the destination container. The pipeline executed successfully, and IAM role assignments ensured secure communication between Azure Data Factory and Azure Storage. Overall, this project strengthened my understanding of Azure cloud fundamentals, data integration, and pipeline orchestration, providing practical experience with industry-standard cloud data engineering tools.